UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Master budgeting

Strategic Management Accounting

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2nd tutorial session

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EXERCISE 1:
Jessi Corporation

EXERCISE 1: Jessi Corporation

The marketing department Of **Jessi Corporation** has submitted the following sales forecast for the upcoming fiscal year (all sales are on account).

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Budgeted unit sales	11,000	12,000	14,000	13,000

The selling price of the company's product is $18.00 per unit. Management expects to collect 65% of sales in the quarter in which the sales are made, 30% in the following quarter, and 5% of sales are expected to be uncollectible. The beginning balance of accounts receivable, all of which is expected to be collected in the first quarter, is $70,200.

The company expects to start the first quarter with 1,650 units in finished goods inventory.

Management desires an ending finished goods inventory in each quarter equal to 15% of the next quarter's budgeted sales. The desired ending finished goods inventory for the fourth quarter is 1,850 units.

EXERCISE 1: Jessi Corporation

Requirement:

1. Calculate the estimated sales for each quarter of the fiscal year and for the year as a whole.
2. Calculate the expected cash collections for each quarter of the fiscal year and for the year as a whole.
3. Calculate the required production in units of finished goods for each quarter of the fiscal year and for the year as a whole.

EXERCISE 1: Jessi Corporation

Q1. Calculate **the estimated sales** for each quarter of the fiscal year and for the year as a whole.

Jessi Corporation Sales Budget				
1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
11,000	12,000	14,000	13,000	50,000
× $18.00	× $18.00	× $18.00	× $18.00	× $18.00
$198,000	$216,000	$252,000	$234,000	$900,000

Estimated sales = Budgeted unit sales x Selling price per unit

EXERCISE 1: Jessi Corporation

Q2. Calculate **the expected cash collections** for each quarter of the fiscal year and for the year as a whole.

Schedule of Expected Cash Collections					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Beginning accounts receivable	$ 70,200				$ 70,200
1st Quarter sales (65%, 30%)	128,700	$ 59,400			188,100
2nd Quarter sales (65%, 30%)		140,400	$ 64,800		205,200
3rd Quarter sales (65%, 30%)			163,800	$ 75,600	239,400
4th Quarter sales (65%)				152,100	152,100
Total cash collections	**$198,900**	**$199,800**	**$228,600**	**$227,700**	**$855,000**

1st Quarter → 198,000 x 0.65 = $ 128,700 in the following 198,000 x 0.30 = $ 59,400

2nd Quarter → 216,000 x 0.65 = $ 140,400 in the following 216,000 x 0.30 = $ 64,800

3rd Quarter → 252,000 x 0.65 = $ 163,800 in the following 252,000 x 0.30 = $ 75,600

4th Quarter → 234,000 x 0.65 = $ 152,100

EXERCISE 1: Jessi Corporation

Q3. Calculate the **required production in units of finished goods** for each quarter of the fiscal year and for the year as a whole.

Jessi Corporation Production Budget					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Budgeted unit sales	11,000	12,000	14,000	13,000	50,000
Add desired units of ending finished goods inventory	1,800	2,100	1,950	1,850	1,850
Total needs	12,800	14,100	15,950	14,850	51,850
Less units of beginning finished goods inventory	1,650	1,800	2,100	1,950	1,650
Required production in units	**11,150**	**12,300**	**13,850**	**12,900**	**50,200**

For end of
1st quarter → 12,000 units × 0.15 = 1,800 units
2nd quarter → 14,000 units × 0.15 = 2,100 units
3rd quarter → 13,000 units × 0.15 = 1,950 units

EXERCISE 2:
Hruska Corporation

EXERCISE 2: Hruska Corporation

The production department of Hruska Corporation has submitted the following forecast of units to produce by quarter for the upcoming fiscal year:

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Units to be produced	12,000	10,000	13,000	14,000

Each unit requires 0.2 direct labor-hours and direct laborers are paid $16.00 per hour.

In addition, the variable manufacturing overhead rate is $1.75 per direct labor-hour. The fixed manufacturing overhead is $86,000 per quarter. The only noncash element of manufacturing overhead is depreciation, which is $23,000 per quarter.

EXERCISE 2: Hruska Corporation

Requirement:

1. Calculate company's total estimated direct labor cost for each quarter of the upcoming fiscal year and for the year as a whole.
2. Calculate the company's total estimated manufacturing overhead cost for each quarter of the upcoming fiscal year and for the year as a whole.
3. Calculate the company's cash disbursement for manufacturing overhead for each quarter of the upcoming fiscal year and for the year as a whole.

EXERCISE 2: Hruska Corporation

Q1. Calculate company's **total estimated direct labor cost** for each quarter of the upcoming fiscal year and for the year as a whole.

Hruska Corporation Direct Labor Budget					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Required production in units	12,000	10,000	13,000	14,000	49,000
Direct labor time per unit (hours)	0.2	0.2	0.2	0.2	0.2
Total direct labor-hours needed	2,400	2,000	2,600	2,800	9,800
Direct labor cost per hour	$16.00	$16.00	$16.00	$16.00	$16.00
Total direct labor cost	**$38,400**	**$32,000**	**$41,600**	**$44,800**	**$156,800**

Total direct labor-hours needed = Required production in units x Direct labor time per unit

Total direct labor cost = Total direct labor-hours needed x Direct labor cost per hour

EXERCISE 2: Hruska Corporation

Q2. Calculate the company's total **estimated manufacturing overhead cost** for each quarter of the upcoming fiscal year and for the year as a whole.

Hruska Corporation Manufacturing Overhead Budget					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Budgeted direct labor-hours	2,400	2,000	2,600	2,800	9,800
Variable manufacturing overhead rate	$1.75	$1.75	$1.75	$1.75	$1.75
Variable manufacturing overhead	$ 4,200	$ 3,500	$ 4,550	$ 4,900	$ 17,150
Fixed manufacturing overhead	86,000	86,000	86,000	86,000	344,000
Total manufacturing overhead	**90,200**	**89,500**	**90,550**	**90,900**	**361,150**

Variable manufacturing overhead = Budgeted direct labor-hours x Variable manufacturing overhead rate

Total estimated manufacturing overhead cost = Variable manufacturing overhead + Fixed manufacturing overhead

EXERCISE 2: Hruska Corporation

Q3. Calculate the company's **cash disbursement for manufacturing overhead** for each quarter of the upcoming fiscal year and for the year as a whole.

Hruska Corporation Manufacturing Overhead Budget					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Budgeted direct labor-hours	2,400	2,000	2,600	2,800	9,800
Variable manufacturing overhead rate	$1.75	$1.75	$1.75	$1.75	$1.75
Variable manufacturing overhead	$ 4,200	$ 3,500	$ 4,550	$ 4,900	$ 17,150
Fixed manufacturing overhead	86,000	86,000	86,000	86,000	344,000
Total manufacturing overhead	90,200	89,500	90,550	90,900	361,150
Less depreciation	23,000	23,000	23,000	23,000	92,000
Cash disbursements for manufacturing overhead	**$67,200**	**$66,500**	**$67,550**	**$67,900**	**$269,150**

Cash disbursement for manufacturing overhead = Total manufacturing overhead - depreciation

EXERCISE 3:
Zan Corporation

EXERCISE 3: Zan Corporation

The production department of Zan Corporation has submitted the following forecast of Units to be produced by quarter for the upcoming fiscal year:

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Units to be produced	5,000	8,000	7,000	6,000

In addition, 6000 grams of raw material inventory on hand at the start of the 1st Quarter and the beginning accounts payable for the 1st Quarter is $2,880.

Each unit requires 8 grams of raw material that costs $1.20 per gram. Management desires to end each quarter with any inventory of raw materials equal to 25% of the following quarter's production needs. The desired ending inventory for the 4th Quarter is 8,000 grams.

Management plans to pay for 60% of raw material purchases in the quarter acquired and 40% in the following quarter.

Each unit requires 0.20 direct labor-hours and direct laborers are paid $15 per hour.

EXERCISE 3: Zan Corporation

Requirement:

1. Calculate estimated grams of raw material that need to be purchased each quarter and for the year as a whole
2. Calculate the cost of raw material purchases for each quarter and for the year as a whole.
3. Calculate the expected cash disbursements for purchases of materials for each quarter and for the year as a whole.
4. Calculate the estimated direct labor cost for each quarter and for the year as a whole.

EXERCISE 3: Zan Corporation

Q1. Calculate **estimated grams of raw material that need to be purchased** each quarter and for the years as a whole.

Zan Corporation Direct Materials Budget					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Required production in units of finished goods	5,000	8,000	7,000	6,000	26,000
Raw materials needed per unit of finished goods	× 8	× 8	× 8	× 8	× 8
Raw materials needed to meet production	**40,000**	**64,000**	**56,000**	**48,000**	**208,000**

EXERCISE 3: Zan Corporation

Q2. Calculate **the cost of raw material purchases** for each quarter and for the year as a whole.

Zan Corporation Direct Materials Budget					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Required production in units of finished goods	5,000	8,000	7,000	6,000	26,000
Raw materials needed per unit of finished goods	× 8	× 8	× 8	× 8	× 8
Raw materials needed to meet production	40,000	64,000	56,000	48,000	208,000
Add desired units of ending raw materials inventory	16,000	14,000	12,000	8,000	8,000
Total units of raw materials needed	56,000	78,000	68,000	56,000	216,000
Less units of beginning raw materials inventory	6,000	16,000	14,000	12,000	6,000
Units of raw materials to be purchased	50,000	62,000	54,000	44,000	210,000
Unit cost of raw materials	× $1.20	× $1.20	× $1.20	× $1.20	× $1.20
Cost of raw materials to be purchased	$60,000	$74,400	$64,800	$52,800	$252,000

desired units of ending raw materials inventory in End of
1st quarter: 64,000 × 25% = 16,000 grams
2nd quarter: 56,000 × 25% = 14,000 grams
3rd quarter: 48,000 × 25% = 12,000 grams

EXERCISE 3: Zan Corporation

Q3. Calculate the **expected cash disbursements for purchases of materials** for each quarter and for the year as a whole.

Zan Corporation					
Schedule of Expected Cash Disbursements for Materials					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Beginning accounts payable	$ 2,880				$ 2,880
1st Quarter purchases	36,000	$24,000			60,000
2nd Quarter purchases		44,640	$29,760		74,400
3rd Quarter purchases			38,880	$25,920	64,800
4th Quarter purchases				31,680	31,680
Total cash disbursements for materials	$38,880	$68,640	$68,640	$57,600	$233,760

1st Quarter → 60,000 x 0.60 = $ 36,000 in the following $ 60,000 x 0.40 = $ 24,000

2nd Quarter → 74,400 x 0.60 = $ 44,640 in the following $ 74,400 x 0.40 = $ 29,760

3rd Quarter → 64,800 x 0.60 = $ 38,880 in the following $ 64,800 x 0.40 = $ 25,920

4th Quarter → 52,800 x 0.60 = $ 31,680

EXERCISE 3: Zan Corporation

Q4. Calculate the **estimated direct labor cost** for each quarter and for the year as a whole.

Zan Corporation Direct Labor Budget					
	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**	**Year**
Required production in units	5,000	8,000	7,000	6,000	26,000
Direct labor-hours per unit	× 0.20	× 0.20	× 0.20	× 0.20	× 0.20
Total direct labor-hours needed	1,000	1,600	1,400	1,200	5,200
Direct labor cost per hour	× $15.00	× $15.00	× $15.00	× $15.00	× $15.00
Total direct labor cost	$ 15,000	$ 24,000	$ 21,000	$ 18,000	$ 78,000